

Prop proyeccion ortogonal convexo cerrado

Proposición 1. Sea H un espacio de Hilbert y $S \subset H$ no vacío, cerrado y convexo

$$\implies \forall x \in H : \exists! y \in S : \|x - y\| = \min\{\|x - z\| : z \in S\}.$$

Denotamos $y = P_S(x)$ y la aplicación $P_S: H \rightarrow S$ se llama **proyección ortogonal**.

Demostración: Repetir la prueba del teo-cerrado-convexo-hilbert-imp-exists-min-norma/??, trasladando el origen a x . ■

Referenciado en

- teo-con-fundamental-iff-sistema-ortogonal-completo
- teo-carac-proyeccion-ortogonal-subespacio-cerrado
- prop-carac-proyeccion-ortogonal-convexo-cerrado
- teo-proyeccion-ortogonal-sistema-ortonormal-formula

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